

A cardinality obstruction to a height-gap value set

A disproof of one assertion of conjecture 00000007766

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Conjecture 00000007766 asserts, among other things, that the value set of a real-valued height difference D on periodic points in $\mathbb{P}^1(\overline{\mathbb{Q}})$ consists of 0 and every real number at least a certain positive constant c . We show that no real-valued function on those periodic points can have such a value set.

The algebraic closure $\overline{\mathbb{Q}}$ is countable: each algebraic number is a root of a nonzero polynomial in $\mathbb{Q}[X]$, there are countably many such polynomials, and each has finitely many roots. Also

$$\mathbb{P}^1(\overline{\mathbb{Q}}) = \{[z : 1] : z \in \overline{\mathbb{Q}}\} \cup \{[1 : 0]\},$$

so this projective line is countable. For any self-map F of it, the set $\text{Per}(F)$ of periodic points is a subset of a countable set. Consequently, for any function $D : \mathbb{P}^1(\overline{\mathbb{Q}}) \rightarrow \mathbb{R}$, the image $D(\text{Per}(F))$ is countable.

On the other hand, $[c, \infty)$ is uncountable for every real c : the map $x \mapsto c + e^x$ injects $\mathbb{R}$ into it. Hence

$$[c, \infty) \not\subseteq D(\text{Per}(F)),$$

and in particular $D(\text{Per}(F)) \neq \{0\} \cup [c, \infty)$. This directly contradicts the conjecture's stated value-set assertion, regardless of the precise formula for D .

Formalization. The Lean file takes $K = \text{AlgebraicClosure}(\mathbb{Q})$ and represents $\mathbb{P}^1(K)$ by `Option K`, with `some z` representing $[z : 1]$ and `none` representing $[1 : 0]$. It proves this type countable from the algebraicity of every element of K . For arbitrary F and arbitrary real-valued D , the theorem `TLMC7766.no_claimed_value_set` rules out the asserted exact value set. Its axiom audit contains only `propext`, `Classical.choice`, and `Quot.sound`.

Scope. The argument does not require the formula for the canonical heights or the dual map. It refutes the universal coverage of a real interval claimed for their difference on algebraic periodic points. It makes no claim about the separate tail asymptotic.